

# PUSHES & PULLS

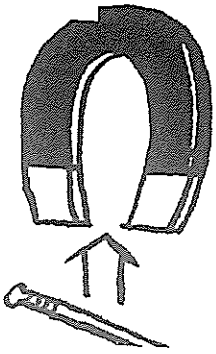
CLASS:

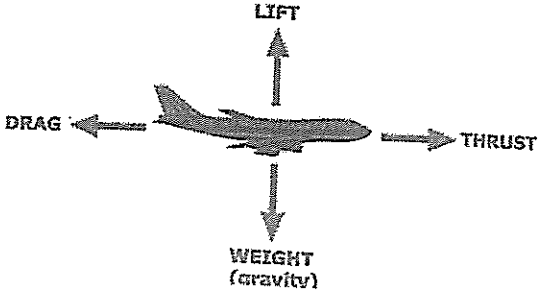
.....

NAME:

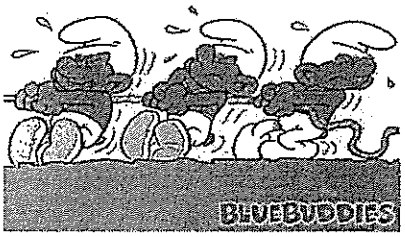
.....





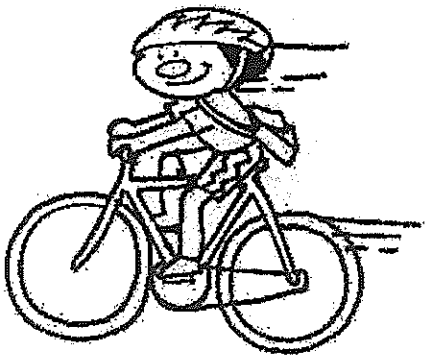


# PUSHES & PULLS




## YOU SHOULD KNOW OR BE ABLE TO DEMONSTRATE THE FOLLOWING

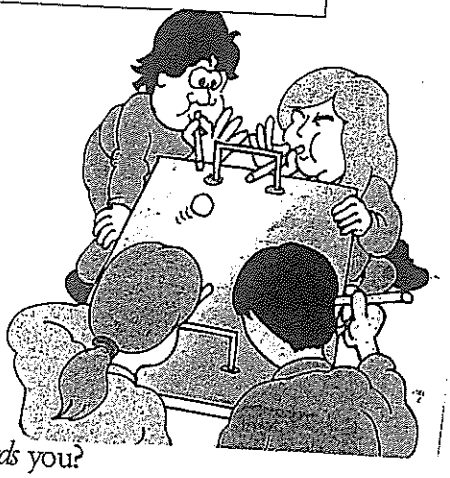
- Know that forces are pushes or pulls
- Know the different types of forces
- Know the effect of forces on an object
- Use arrows to show the size and direction of a force
- Know what contact and non-contact forces are and give examples
- Know that forces may be balanced or unbalanced
- Know that if forces are balanced, an object remains stationary
- Know that if forces are unbalanced, the object will start moving, speed up or change direction or shape
- Know that forces are measured in newtons (N) using Newton balances
- Know that friction is a contact force that occurs when two things rub against each other and that it slows down or prevents motion
- Know that friction depends on the type of surface and mass of the object
- Know the different ways of reducing friction
- Know that gravity is the force of attraction between any two objects e.g. between the Sun and a planet; Earth and an object. Gravity depends on the masses of the objects and the distance between them
- Know the difference between weight and gravity. Weight is the downward pull of gravity on the object. Mass is the amount of matter in it.



# EXPERIMENT 1: EXPERIENCING FORCES

There are seven stations set out in the lab. Perform each activity and answer the relevant questions before you move on to the next station. May the force be with you!

## PART A: What will a push do to a moving object?



Use the table tennis ball and a straw to answer these questions.

a) What happens if you blow on the ball while it is moving *towards* you?

.....

b) What happens if you blow on it while it is moving *away* from you?

.....

c) What happens if you blow on it while it is moving *across* in front of you?

.....

## PART B: Which object will drop faster?

Hold the piece of aluminium in one hand and the piece of lead in the other, at the same height. Drop them both at the same time?

a) Predict which one will drop faster? The one will less or more mass?

.....

b) Which one actually dropped faster or did both drop at the same speed?

.....

c) What force is pulling the objects down?

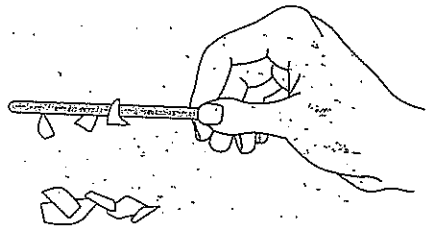
.....

d) Will these objects fall at the same speed on the Moon? Explain your answer.

.....

.....

## PART C: Can rubbing cause an electric charge?



Rub the plastic rod with the woolen cloth for 1 minute. Hold it near some small bits of paper (without toching them).

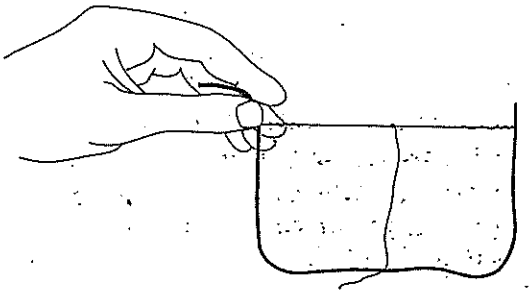
a) What happened to the bits of paper?

.....

b) What force is involved in pulling the paper to the rod?

.....

## PART D: What is surface tension?



Dip the U-shaped frame into some soap solution. A soap film should form on the frame. Pull the thread a stretch the soap film without breaking it. Then release the thread. Hold the frame and let the thread hang over the frame as shown. *Break* the film on one side with something sharp.

a) What happened to the string when the film was broken?

.....

b) What force is moving the thread?

.....

## PART E: Moving magnets

Hold one magnet in each hand. Feel what happenes as you slowly bring the end of one magnet close to the end of the other.

Repeat the test, but use the other end of *one* of the magnets.

a) What happened to the magets?

.....

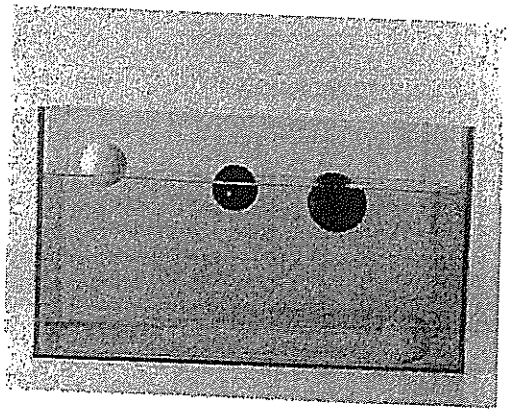
b) What happened when you repeated it with the other end of one of the magets

.....

c) Write a generalisation describing the forces between the two magnets.

.....

## PART F: What is upthrust?



Put a table-tennis ball into a bucket of water. Push it to the bottom of the bucket and let it go. Repeat with a tennis ball, rubber and styrofoam balls.

a) What happened to the balls?

.....

b) Which ball popped up faster? Why?

.....

c) Write a generalisation about what happens to objects in water.

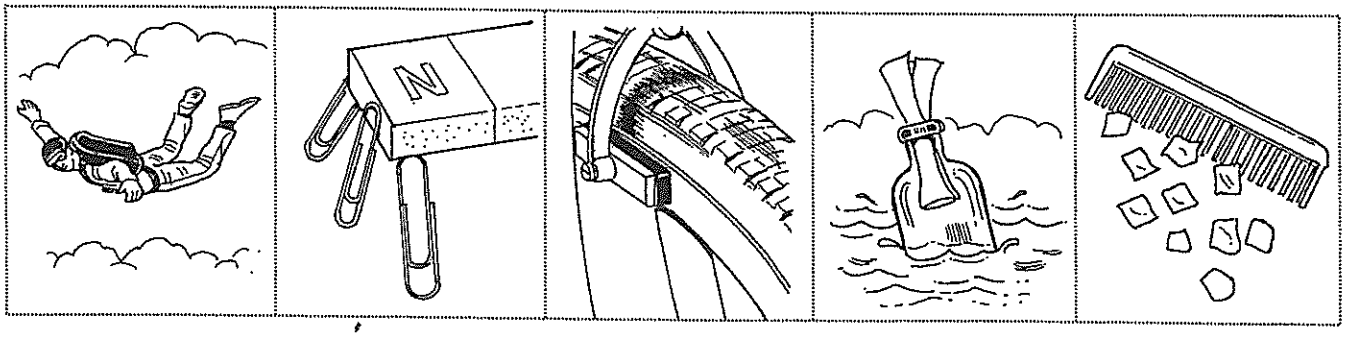
.....

# Contact and non-contact forces

1 Cut out the five pictures and five labels at the bottom of the page.
2 Stick the forces in the correct boxes.
3 Stick the correct label below each picture.

**Non contact forces**

**Contact forces**




| Magnetism | Static electricity | Friction | Gravity | Upthrust |
|---|---|---|---|---|

**[ knowledge ]**

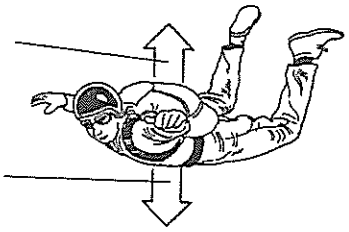
# What do forces do?

1 Look at the four pictures below. The size of the arrow represents the size of the force and its direction.

For each picture fill in the gaps.

**A** This force is called _____

This force is called _____

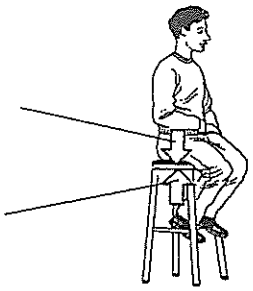


The forces are _____

(balanced/unbalanced).

**B** This force is called _____

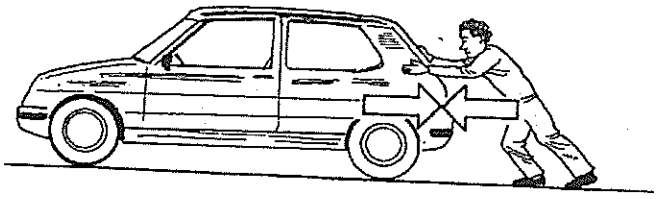
This force is called _____



The forces are _____

(balanced/unbalanced).

**C**



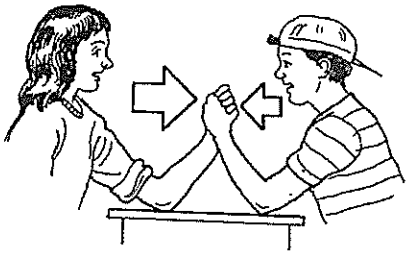
The forces are _____

(balanced/unbalanced).

The car _____

(will/will not) move.

**D**



The forces are _____

(balanced/unbalanced).

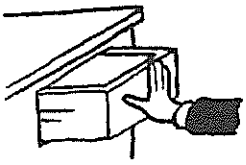
The _____ (boy/girl)

will win the contest.

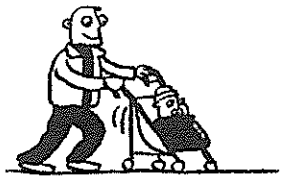
**[ knowledge ]**

# Forces are everywhere

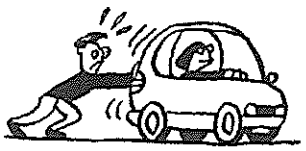
**1** Which forces are pushes and which ones are pulls? Write the correct words in the spaces.



_____



_____



_____



_____

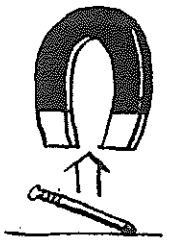
**2** Look at the pictures below. Write the names of the forces indicated by the arrow.

**A**



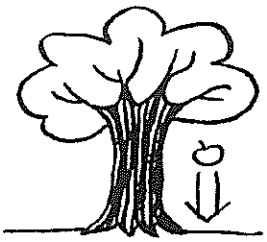
_____

**B**



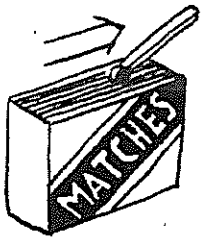
_____

**C**



_____

**D**



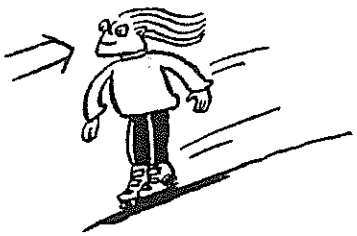

_____

**E**



_____

**F**



_____

**3** Which pictures show non-contact forces? _____

**4** Complete these sentences.

**a** A _____ is needed to start an object moving.

**b** A force can change the _____, _____ and _____ of an object.

**c** Nothing starts to move without a _____ being involved.

**d** Three forces which can act from a distance are _____, _____ and _____ _____ forces.

**[ knowledge ]**

# Find the forces

Write the answers to the clues in the spaces.

| Clues | | Answers |
|---|---|---|
| **a** | A force which pulls you down. | _____ |
| **b** | This force helps to hold things to fridge doors. | _____ |
| **c** | This type of force needs to touch something to affect it. | _____ |
| **d** | This force rubs things away. | _____ |
| **e** | This force helps a ship float. | _____ |
| **f** | A form of electricity which can attract things. | _____ |

Now try to find the words in the wordsearch. Mark all the contact forces in red and the non-contact forces in blue.

(Use pen and pencil if you do not have any coloured pencils.)

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| W | F | R | I | C | T | I | O | N |
| S | R | A | O | P | D | Y | Q | L |
| A | F | C | H | R | T | Y | U | M |
| S | T | E | O | I | U | F | S | J |
| I | S | I | V | N | P | I | T | P |
| D | V | A | R | E | T | I | O | M |
| B | R | N | I | E | H | A | I | E |
| G | I | K | N | O | R | U | C | L |
| R | C | G | D | I | U | E | M | T |
| S | A | I | B | O | S | K | F | U |
| M | F | S | T | A | T | I | C | Z |

**[ knowledge, literacy ]**

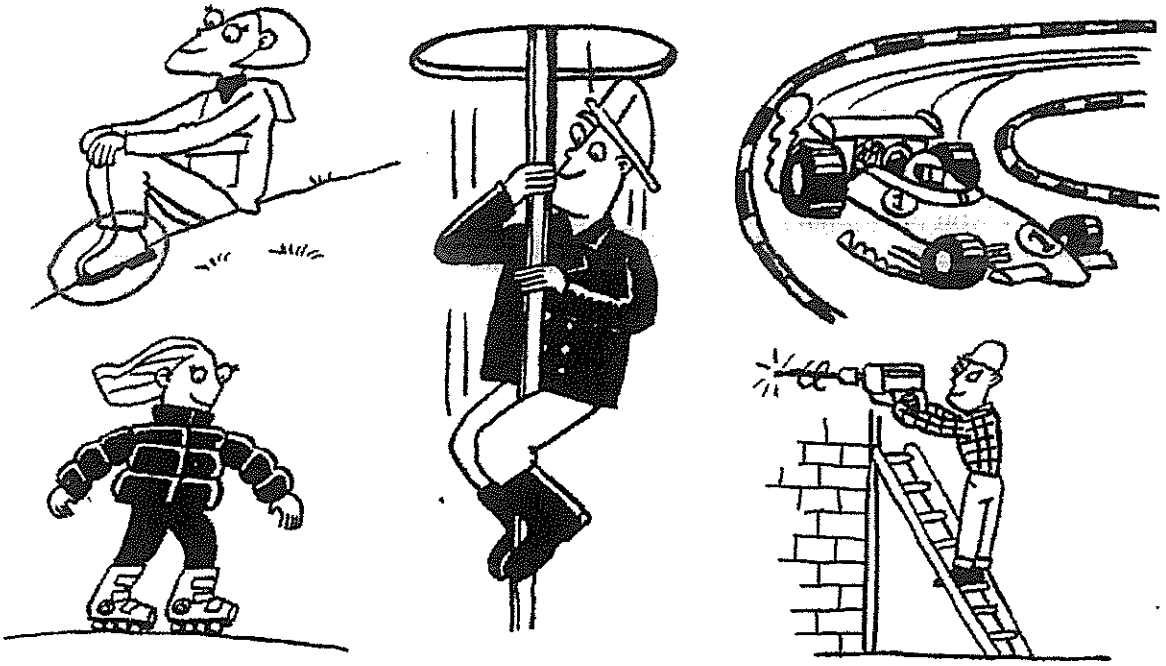
# Friction around us

1 Copy the sentences and fill in the missing words. The words you need are in the box.

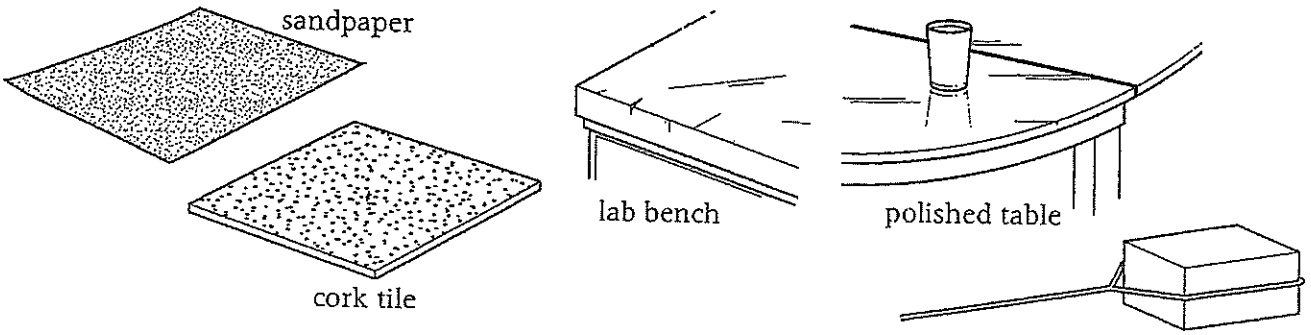
| force | rough | smooth | surfaces |
|---|---|---|---|

Friction is a _____ caused by two _____ moving over each other. _____ surfaces produce more friction than _____ surfaces.

2 Look at these diagrams.
Put a circle around the all the places where there is friction. One has been done for you.



3 Look at these pictures.




You pull the block along all of these surfaces.

**a** It will be easiest to pull the block along the _____ .

This is because it has the _____ (smoothest/roughest)

surface, so there will be the _____ (most/least) friction.

**b** It will be hardest to pull the block along the _____.

This is because it has the _____ (smoothest/roughest) surface, so there

will be the _____ (most/least) friction.

**[ knowledge ]**

# EXPERIMENT 2: MEASURING FRICTION

Carry out the experiment and complete the report.

**Aim:**

.....

.....

**Prediction:**

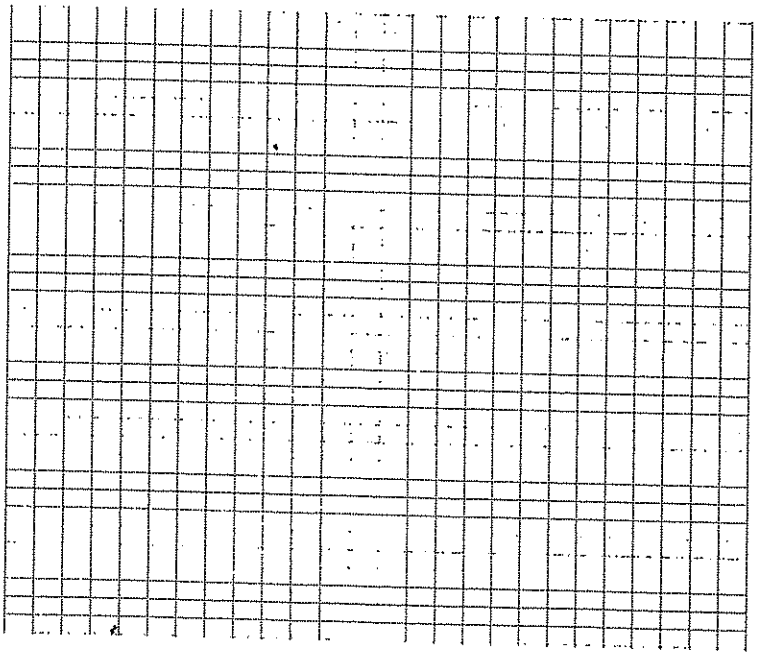
.....

.....

**Results:**

| | Force needed to pull block (N) | | | |
|---|---|---|---|---|
| Surface tested | Trial 1 | Trial 2 | Trial 3 | Average |
| Desk top | | | | |
| Carpet | | | | |
| Lino | | | | |
| Sandpaper | | | | |
| Concrete path | | | | |

Draw a bar graph of your data. Make sure to give your graph a title and label your axes including units of measurement.



**Conclusion** (Which surface produced the most friction?) Which produced the least?)

.....

.....

.....

**Discussion** (Use scientific knowledge to explain your results)

.....

.....

.....

.....

.....

**Evaluation:**

(Why was it necessary to take 3 measurements each time, instead of just 1?)

.....

.....

.....

(Was it a *fair test*? How can you make it a fair test?)

.....

.....

.....

(State any problems that you encountered in doing the experiment)

.....

.....

.....

(Suggest any improvements you can make to the experiment)

.....

.....

.....

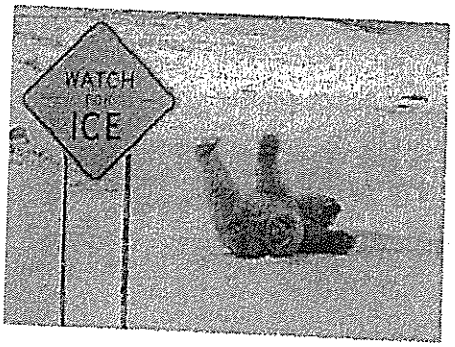
# Friction – true or false?

The statements below are all about friction. Against each of them indicate if it is **True, False**, or **Partly True.**

| | |
|---|---|
| Friction always slows things down. | Cars need friction to keep moving. |
| Cars need friction to stop. | You could not walk without friction. |
| Friction is useful to gymnasts. | Matches light because of friction. |
| Friction is useful to ships. | Shoelaces stay tied up because of friction. |
| You could not pick up a cup of tea without friction. | You could drink from a glass without friction. |
| Snow increases the friction between your shoes and the ground. | Friction is useful in playgrounds. |
| Pencils do not need friction to write. | There is no friction when you are roller skating. |

**[ knowledge ]**

# FRICTION

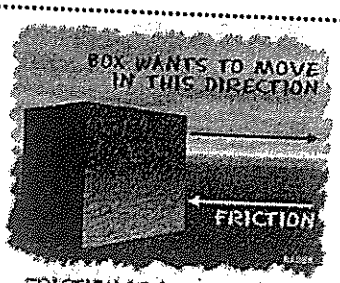



1. Give 3 examples when friction is a nuisance
   .....
   .....
   .....

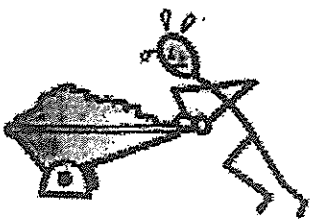
2. Give 3 examples when friction is useful
   .....
   .....
   .....
   .....

3. Think about why you slip on a wet floor, why oil oil is used in a car engine, why you use grease on a bicycle chain, why you put soap on your finger to remove a tight ring, and why your mouth produces saliva.
   Water, oil, grease, soap and saliva are all substances that
   .....
   These substances are called .....

4. Lindy and Julie are having a race on a dry water slide. Lindy is sliding on a smooth mat and Julie is using roller skates. Who do you think will reach the bottom first? Why? Explain your answer in terms of reducing friction.
   .....
   .....
   .....
   .....
   .....
   .....

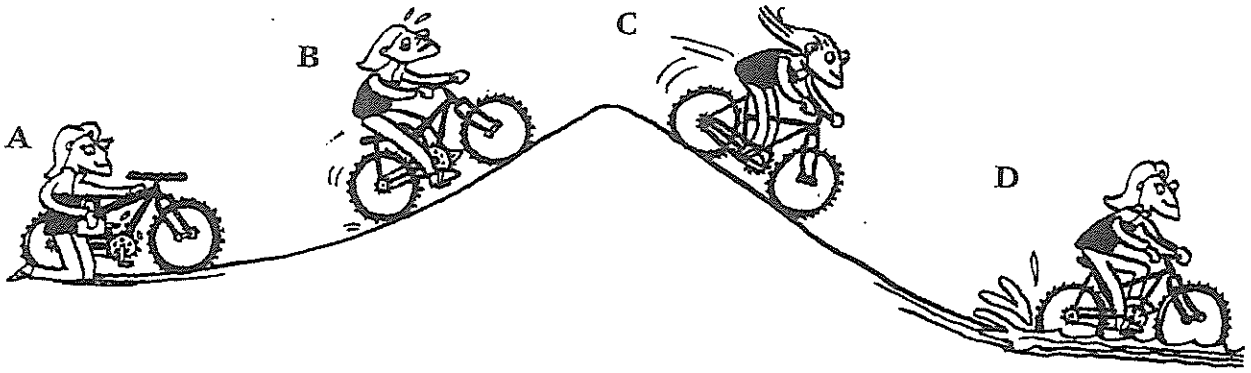



FRICTION IS A FORCE THAT MOVES IN AN OPPOSITE DIRECTION TO MOVEMENT



Only flat on the bottom! Ha Ha!

# Lubrication




1 **a** Which parts of the bike should the girl oil? Explain why.

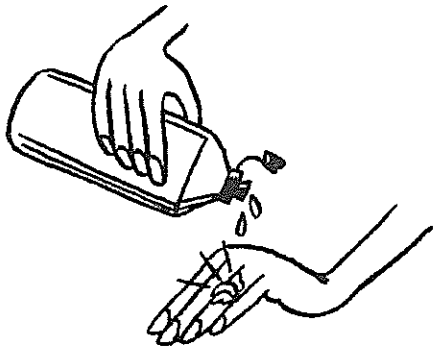
**b** You should never put oil on the wheel rim where the brake blocks touch it. Explain why not.

**c** The girl going uphill is working very hard. What forces is she pushing against?

**d** There are two forces slowing down bike C. What are they?

**e** What will happen to bike D as it goes through the puddle? Why will this happen?

Anita got her mum's wedding ring stuck on her finger. Her mum added a little washing up liquid to the ring before trying to pull it off.



2 **a** How can adding washing up liquid help remove the ring?

**b** The washing up liquid acted as a _____.

**c** What other substances could be used instead of washing up liquid? Give two examples.

Angela enjoys going fast on her skis. Before going on the slope, she always rubs wax on the bottom of her skis.



3 **a** How does the wax help Angela ski faster?

**b** What two forces slow her down as she travels down a ski slope?

**[ knowledge ]**

# Gravity questions

1 On the Earth, gravity pulls with a force of 10 N on every kilogram of mass.
To find the weight of a particular mass, multiply the mass by 10. If you know the weight and want to find the mass, divide by 10.
Fill in the gaps in this table.

| Object | Mass (kg) | Weight (N) |
|---|---|---|
| television | 8.0 | |
| radio | 1.5 | |
| chair | | 50 |
| bag of sugar | 1.0 | |
| bottle of lemonade | | 20 |

2 If we are measuring small masses, we usually use grams instead of kilograms. If we have a mass in grams, we need to convert it to kilograms before we find the weight. To change a mass in grams into kilograms, divide by 1000.
Fill in the gaps in this table. The first one is done for you.

| Object | Mass (g) | Mass (kg) | Weight (N) |
|---|---|---|---|
| jar of jam | 340 | 0.34 | 3.4 |
| bag of pasta | 500 | | |
| tin of beans | 450 | | |
| loaf of bread | 800 | | |
| can of lemonade | 330 | | |

3 Why don't people in Australia feel like they are upside down?



_____

_____

_____

4 The person on the top of the Earth is holding out a sports bag.
Draw in a sports bag for the other person.

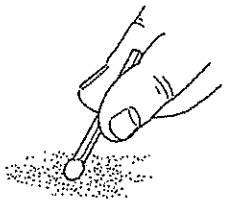


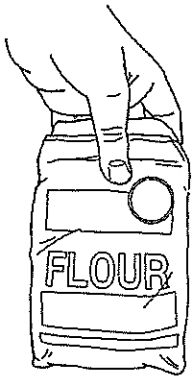
**[ knowledge, numeracy ]**

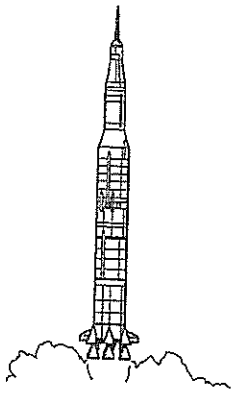
# Measuring forces

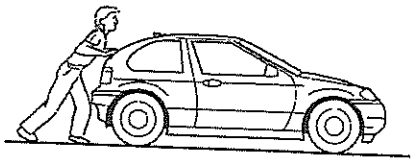
1 The unit of force is called a _____ .

2 The pictures below show examples of forces. Write the correct size of each force in the space below each picture. Choose numbers from the box.







_____ _____ _____ _____

| |
|---|
| **50 000 000 N** |
| **0.5 N** |
| **650 N** |
| **10 N** |

3 Fill in the missing words:

The Earth _____ all objects towards it. This force is called _____ . It is this force which gives us _____ .

4 Complete these sentences using words from the box:

The weight of a 2 kg mass on Earth is _____ .

The typical mass of a man on Earth is about _____ .

The unit of mass is the _____ .

The unit of force is the _____ .

The mass of a 2 kg bag of apples on the Moon is _____ .

| | | | | |
|---|---|---|---|---|
| **kilogram** | **newton** | **20 N** | **2 kg** | **80 kg** |

**[ knowledge ]**

# FORCES REVISION SHEET

## *Fill in the blanks*

1. A force can be thought of as a push or a pull. The units of force are _____ ( ). Larger forces may be measured in kilonewtons (kN). 1 kilonewton = 1000 newtons.

2. Forces can have several effects. They may:

   A. Keep a body _____ ('at rest').

   B. Cause a body which is at rest to begin to _____.

   C. Change the _____ of a moving body (make it accelerate or decelerate).

   D. Change the _____ of a body (squash, stretch or bend it).

   E. Change the _____ of a moving body.

   F. Cause a body which is moving to come to a _____.

3. Look at the pictures below.
   Which of the effects from Q.2 are used in each of these?

1.



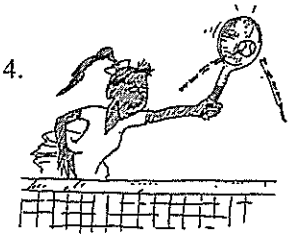
2.



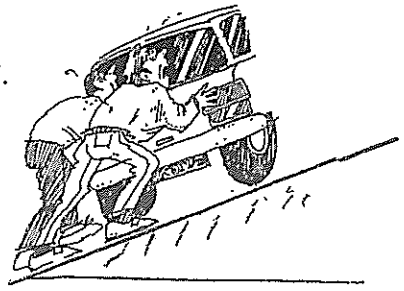
3.



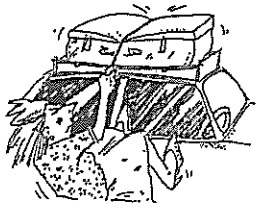
4.



5.



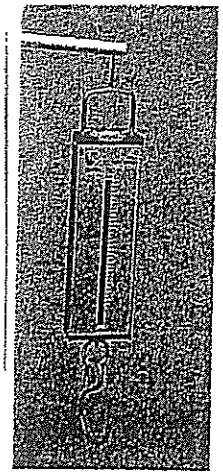
6.



4. The picture shows

a _____.

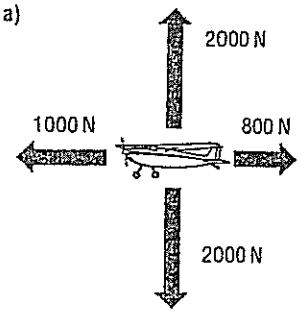
It is used to measure _____.



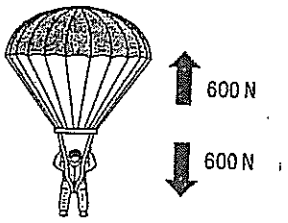
5. If the forces on a moving object are not balanced the object will speed up or slow down. A stationary object will not move because the forces on the object are _____.

6. What effect will the following forces have on the motion of the object?

a)




b)



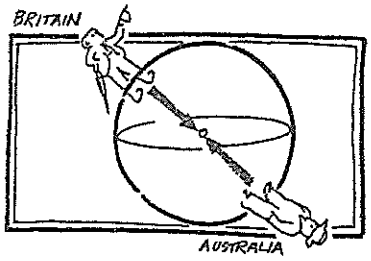

c)

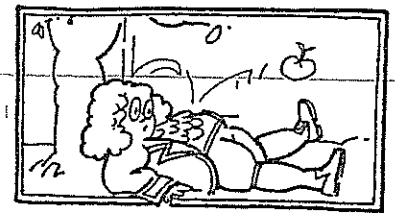



d) moving already



7. *Weight* is a very common force. Weight is the force of gravity due to the pull of the _____. This is the force that is pulling you downwards now. It acts towards the _____ of the Earth.



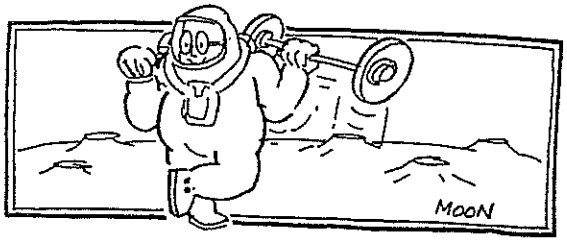

8. 

An apple fell on Newton's head,
Moved downwards from the tree,
And that was when Sir Isaac said:
"The force is _____."

9. The weight of an object varies slightly at different places on the Earth.
It is _____ on top of a mountain than at sea level since it is _____ from the centre of the Earth.




10. If an object is taken to the Moon, it weights only about _____ as much because the Moon is smaller than the Earth.
In outer space, well away from any planets, objects become _____.



11. The **mass** of an object is a measure of its inertia – the resistance against any attempt to change its motion. The mass depends on the amount of _____ in the object and is measured in _____. At the earth's surface, gravity pulls on each kilogram of matter with a force of 10 newtons (10N). The value (10 newtons per kilogram) is the **gravitational field strength** of the earth.

To change mass into

weight (N) = mass (kg) x gravitational field strength (N/kg)

On earth, the gravitational field strength = 10 N/kg.

So:

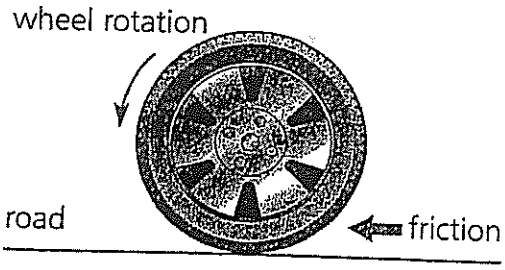
weight (N) = mass (kg) x ____

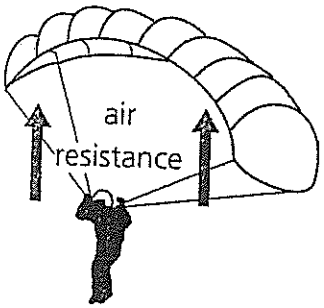
12.

The table below shows the gravitational field strength at the surface of some of the planets and the moon.
Complete the gaps in the table.

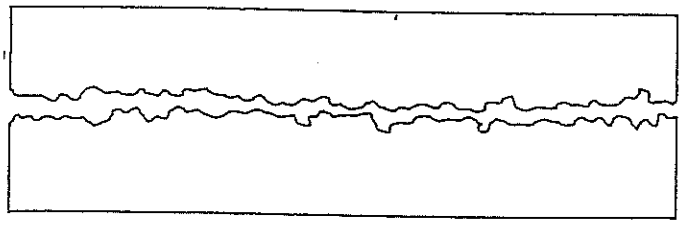
| Planet | Gravitational field strength (N/kg) | Mass of object on planet's surface (kg) | Weight of object (N) |
|---|---|---|---|
| Earth | 10 | 20 | |
| Moon | 1.66 | | 1.66 |
| Venus | | 2 | 18 |
| Mars | 4 | | 16 |
| Jupiter | 26 | | 104 |
| Neptune | 14 | 10 | |

13. **Friction** is a force that tries to prevent _____.
Friction can sometimes be an advantage and sometimes it can be a disadvantage. Friction between a car tyre and the road is vital for the car to move and for it to _____ down without skidding.
The more tread there is on a tyre the greater the friction. There are very strict rules about how much tread there should be on a tyre. If there is too little, it would be difficult to stop the car quickly.
A parachutist slows down because of the _____ resistance.



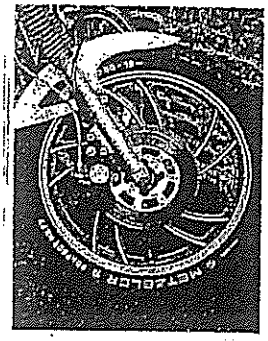

14. Friction is caused by the _____ of surfaces. all surfaces are really quite rough if we look at them under a microscope, even those which we think are very flat and smooth.



If two 'flat' surfaces are in contact, these humps and hollows will fit together. So when one surface tries to slide over the other they tend to stick. This is what causes friction.

15. We live in a world where we experience the effects of friction all the time. Friction between the soles of our shoes and the _____ enables us to walk. If there is ice on the ground, friction is much _____ (though it doesn't disappear entirely) and we have a lot more difficulty moving around!).

16. The material used to make a motorcycle brake-pads is chosen because its surface is hard and _____ and so provides a _____ of friction with a metal disc on the motorcycle wheel.



17. We can reduce friction by preventing surfaces interlocking. The simplest solution is to make the two surfaces roll over each other, rather than sliding. It is easier to move something heavy, such as a cooker, by putting it on _____, instead of dragging it along.

If we must have sliding contact, as in a hinge for example, oil is used to _____ the movement. The oil fills the hollows on the two surfaces, and prevents them fitting together like a jigsaw. This makes it easier for them to slide over one another.

18. One form of transport uses _____ to reduce friction. The hovercraft floats on a cushion of air. This keeps it completely _____ the ground surface and friction is much less.

